

Curriculum Vitae

Politecnico di Milano · PhD in Information Technology, 42nd cycle · call 5722-5460
Auditing and Mitigating Epistemic Failures in Web-Connected AI Agents



PERSONAL INFORMATION

Surname	Ikram
Name	Asad
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Nationality	Pakistani
Date of birth	20 January 1998

EDUCATION AND TRAINING

Date (from – to)	September 2024 – December 2025
Organisation	University of Portsmouth, United Kingdom (public university)
Duration of the programme	1 year full time (180 UK credits; transcript states 90 ECTS)
Principal subjects	Data Analytics. Intelligent Data and Text Analytics; Big Data and Machine Learning; Explainable AI; Data Management (80/100)
Title of qualification	MSc Data Analytics
Final mark obtained	Distinction. UK scale 0–100, where Distinction is the highest classification and requires 70 or above. Masters project mark 78/100.

Date (from – to)	August 2017 – December 2021 (degree awarded April 2022)
Organisation	National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan
Duration of the programme	4 years full time (130 credit hours)
Principal subjects	Computer Science. Data Structures; Database Systems; Operating Systems; Computer Networks; Artificial Intelligence; Applied Machine Learning; Data Mining; Design and Analysis of Algorithms; Probability and Statistics
Title of qualification	BS (Computer Science)
Final mark obtained	CGPA 2.75 on a 0.00–4.00 scale (minimum required for the degree: 2.00). 130 of 130 credits earned; degree status Completed.

GRADUATION THESIS

Title	Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies
Language	English
Supervisor	Dr Ella Haig, Associate Professor in Artificial Intelligence, University of Portsmouth
Summary	This thesis investigated whether social-media platforms can be treated as equivalent sources when measuring public sentiment toward UK economic policies. I collected ~279,000 public posts across X/Twitter, Reddit, YouTube and Quora using official APIs and licensed feeds under ethics approval and GDPR-compliant procedures. A fine-tuned BERT classifier achieved macro-F1 0.878, with human validation at Cohen's kappa ~0.78, temperature-scaling calibration and statistical significance testing. NRC emotion analysis, LIME interpretability and Prophet forecasting were also applied. The central finding was that the same policy event produced structurally different sentiment and emotion profiles across platforms.

PUBLICATIONS AND ARTICLES SUBMITTED

Author(s) and title	Ikram, A. and Haig, E. — “Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment”
Language	
Publication place	Manuscript (complete; not published).
Date of publication	2026 · manuscript completed; not submitted.

CERTIFICATIONS

GRE	Not taken
GMAT	Not taken
Language certification	IELTS Academic , test date 28 March 2024, Test Report Form IELTS Academic, test date 28 March 2024, Test Report Form 23PK507881IKRA011A. Overall Band Score 6.5 (Listening 7.0, Reading 6.5, Writing 6.0, Speaking 7.0). CEFR level B2. This meets the minimum overall score of 6.0 required by the call.

WORK EXPERIENCE

Date (from – to)	February 2025 – present
Firm	ArtemisAI Ltd (trading as ViralData), United Kingdom — remote
Sector	Social media intelligence software
Type of employment	Founder and Chief Technology Officer
Main activities	• Founded and lead the company; built the platform end to end, from Meta Graph API ingestion of 30+ pages daily into a layered lakehouse and Redshift warehouse. • Designed and deployed six production NLP model endpoints on AWS SageMaker (sentiment, emotion, toxicity, topic, language, intent), fine-tuned from XLM-R, DistilBERT, DeBERTa-v3 and Detoxify bases on a ~20,000-item weighted dataset, with a per-sample loss weighted by label confidence, class balance and label source. • Built the labelling pipeline these models learn from: an LLM labels a stratified seed set with its reasoning, decision rules are mined from that reasoning into a prompt for a second model, and low-confidence cases route back for review. • Built multimodal analysis into production: CLIP image-text embeddings feeding a trained fusion classifier (macro-F1 0.93 on a gold set), a hybrid OCR pipeline (token-F1 0.928), and video intelligence with speech transcription. • Instrumented the measurement layer itself: a three-tier fallback cascade with logged confidence routing and weekly divergence monitoring. Its measured inter-model agreement (82.9% on sentiment, 0.0% on toxicity) is what motivates my research interest in annotation reliability.
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Date (from – to)	December 2024 – present
Firm	M+C Saatchi Fluency, United Kingdom — remote
Sector	Data consultancy, government and commercial communications
Type of employment	Data Engineer, consultant
Main activities	• Designed and operate the multi-platform social listening pipeline behind campaign intelligence for UK Government departments and for Amazon, Ford, Nike and Reckitt: Sprinklr and Brandwatch feeds plus custom crawlers into AWS (ECS Fargate, Step Functions, Glue, Athena), with SageMaker NLP sentiment models. • Run a fleet of 68 collectors (62 active) with an automated repair agent operating as a named CI workflow, including anti-bot handling, proxy rotation and TLS

fingerprint management, at 99%+ pipeline uptime.

- Maintain data governance on government-facing datasets: documented filter rules, permanent removal lists, pre-delete backups and weekly refresh cycles.

Date (from – to)	February 2022 – August 2024
Firm	Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Pakistan — on site
Sector	Online classifieds and marketplaces
Type of employment	Senior Data Engineer
Main activities	• Led a three-engineer team owning 500+ regional web collectors across 15 MENA countries on Kubernetes/EKS, gathering multilingual platform data in English, Arabic, Urdu and Hindi. • Redesigned the layered data warehouse (RDL, ODL, ADL), improving ETL/ELT throughput by 60%. • Promoted twice in two years: Associate Software Engineer to Data Engineer to Senior Data Engineer.

Date (from – to)	May 2023 – present (part time)
Firm	Fix.com, Canada — remote
Type of employment	Data Engineer, consultant
Main activities	Designed Scrapy/Scrapyd pipelines processing 50M+ data points from 200+ sources; cut crawl time 40% through distributed cluster tuning; introduced Great Expectations validation with automated schema-drift detection, reducing bad production data by 70%.

Other positions	• Lead Data Engineer, VendueTech, Ireland, remote contract, June – July 2025. Led three engineers building real-time auction data ingestion; migrated Azure to AWS. • Data Engineer, consultant, Prefe, Italy, remote, September – December 2024. Rebuilt legacy Scrapy architecture, reducing failure rate 60%; mentored two junior engineers. • Freelance Data Engineer, CXG, Dubai, remote, December 2025 – January 2026. Production crawlers for global luxury fashion brands at 99%+ completeness against advanced anti-bot protections. • Senior Freelance Data Engineer (Level 2 seller, 300+ projects), worldwide, December 2018 – February 2026. 300+ five-star data engineering projects across 400+ websites for 50+ international clients.
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PERSONAL SKILLS AND COMPETENCES

Mother tongue	Urdu
Other language — English	Reading: excellent · Writing: good · Speaking: excellent <i>Stated to match the IELTS profile above rather than above it (Writing 6.0).</i>
Social skills	Worked across Pakistan, the UK, Ireland, Italy, Canada and the UAE, collaborating in multilingual environments involving Urdu, English, Arabic and Hindi data. Chevening Scholar selected from 70,000+ applicants across 160 countries.
Organisational skills	Led three-engineer teams at Dubizzle Labs and VendueTech while founding and running ArtemisAI with a named delivery team. Promoted twice in two years at Dubizzle Labs, progressing from Associate Software Engineer to Data Engineer to Senior Data Engineer.
Other skills	Author and maintainer of web-scraping-guide.com, a free 24-section public engineering reference covering 60+ libraries and tools, anti-bot systems, legal and ethical constraints, production architectures and data quality.

TECHNICAL SKILLS AND COMPETENCES

Machine learning and NLP	Transformer fine-tuning (BERT, XLM-R, DistilBERT, DeBERTa-v3); CLIP multimodal embeddings; Whisper speech transcription; OCR pipelines; BiLSTM; measurement
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and psychometric methods (generalizability theory, measurement invariance, differential item functioning, score linking); calibration; statistical testing (chi-square, McNemar, bootstrap intervals); LIME and attribution; NRC and VADER lexicons; Prophet time-series forecasting.

Data collection

Scrapy, Playwright, Selenium; official platform APIs (Meta Graph, Reddit, YouTube); licensed feeds; proxy and anti-blocking management; TLS fingerprinting; detection anatomy; agentic browsers.

Engineering

Python, PySpark, SQL; AWS (Fargate, Glue, Athena, Redshift, SageMaker, Step Functions, Lambda, DynamoDB); Airflow; Docker and Kubernetes; PostgreSQL; CI/CD with automated quality gates.

ADDITIONAL INFORMATION

Awards

Chevening Scholar 2024/25 — the UK Foreign, Commonwealth and Development Office's flagship global scholarship, awarded to fund the MSc at Portsmouth.

Public work

Author, web-scraping-guide.com — a free public engineering reference on web data collection: 24 sections covering 60+ libraries and tools, six anti-bot systems, legal and ethical constraints, production architectures and data quality.

Artistic skills

Not applicable.

Other skills

See Personal Skills and Competences; author and maintainer of web-scraping-guide.com.

Annexes

None.